kinetic

KINETIC motion controller + vibration puck: manufacturing manual

> **Status: design only; nothing built, powered or run on a motor.** Controller PCB: **order-ready: every automatic check passes**. Vibration puck PCB: **order-ready: every automatic check passes**. AI model: trained on synthetic data only. Firmware: motion/vibration core host-tested; ESP32 firmware compiles for the ESP32-S3 (arduino-cli, core 2.0.17, 0 warnings; `tools/esp32_build.sh`, binaries in `firmware/release/`), not yet run on hardware.

1. What is made

A 112 x 80 mm controller (ESP32-S3, three StepStick driver sockets, 24 V input, 5 V buck, endstops, host UART/USB) and a 25 x 26 mm ADXL345 vibration puck on a 6-pin SPI cable. Drivers (A4988 / DRV8825 / TMC2209 modules) and motors are bought separately.

2. Deliverables

Item	Where	State
Controller PCB (Gerbers, BOM, CPL, KiCad, STEP)	`utility10/pcb/release/kinetic_ctl/` and `kinetic/hardware/kinetic_ctl/`	**order-ready: every automatic check passes**
Vibration puck PCB	`utility10/pcb/release/kinetic_vib/` and `kinetic/hardware/kinetic_vib/`	**order-ready: every automatic check passes**
Motion + protocol library and tests	`kinetic/firmware/lib/kn_motion.[ch]`, `kinetic/firmware/test/test_kn.c`	`make test` passes (370 checks)
Vibration model	`kinetic/sim/vib.py`, `kinetic/firmware/lib/kn_vib*.{c,h}`	synthetic data; C matches Python
ESP32 skeleton	`kinetic/firmware/src/main_kinetic.cpp`	compiles for the ESP32-S3 (0 warnings), not run on hardware
Enclosure STL (printed)	`utility10/pcb/release/kinetic_ctl/4_enclosure/`	generated from the board outline; unprototyped
Docs	`kinetic/docs/` (CONCEPT, DESIGN, PROTOCOL, TEST_PLAN, economics, vib_results)	

3. Controller bill of materials

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
J5	24 V in	TerminalBlock_bornier-2_P5.08mm	KF128-5.08-2P-AA	C474952	hand/THT
D3	SMBJ28A	D_SMB	Littelfuse SMBJ28A	C178227	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
Q5	AO3401A	SOT-23	AO3401A	C15127	SMT (JLC)
D4	BZT52C12	D_SOD-123	DIODES BZT52C12-7-F	C124196	SMT (JLC)
R40	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
C30	100u/35V	CP_Elec_6.3x7.7	Lelon VZH101M1VTR-0607	C176683	SMT (JLC)
C31	10u/50V	C_1210_3225Metric	Samsung CL32A106KBJNNNE	C380537	SMT (JLC)
U2	AP63205WU-7	TSOT-23-6	AP63205WU-7	C2071056	SMT (JLC)
R41	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
C32	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
L1	4.7u	L_Changjiang_FNR5040S	Changjiang FNR5040S4R7MT	C167963	SMT (JLC)
C33	22u/10V	C_0805_2012Metric	Samsung CL21A226MPQNNNE	C29277	SMT (JLC)
C34	22u/10V	C_0805_2012Metric	Samsung CL21A226MPQNNNE	C29277	SMT (JLC)
D5	SS14	D_SMA	SS14	C84343	SMT (JLC)
U3	AP2112K-3.3	SOT-23-5	AP2112K-3.3TRG1	C51118	SMT (JLC)
C35	1u	C_0603_1608Metric	0603 1uF X7R	C15849	SMT (JLC)
C36	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
U1	ESP32-S3-WROOM-1-N8	ESP32-S3-WROOM-1	ESP32-S3-WROOM-1-N8	C2913198	SMT (JLC)
C37	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
C38	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
R42	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
C39	1u	C_0603_1608Metric	0603 1uF X7R	C15849	SMT (JLC)
SW2	Reset (EN)	SW_SPST_B3U-1000P	Omron B3U-1000P	C231329	SMT (JLC)
SW3	Boot (IO0)	SW_SPST_B3U-1000P	Omron B3U-1000P	C231329	SMT (JLC)
J1	USB-C	USB_C_Receptacle_HRO_TYPE-C-31-M-12	TYPE-C-31-M-12	C165948	SMT (JLC)
R43	5.1k	R_0603_1608Metric	0603 5.1k 1%	C23186	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
R44	5.1k	R_0603_1608Metric	0603 5.1k 1%	C23186	SMT (JLC)
D6	Green	LED_0603_1608Metric	Hubei KENTO KT-0603YG yellow-green 570 nm, Vf about 2 V (lights well from 3.3 V with 1 k; the Everlight 19-217 green was out of stock)	C2289	SMT (JLC)
R45	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
R46	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
R47	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
C40	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
J7	Host UART (Pi/PC)	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J3	Vibration sensor SPI	PinHeader_2x03_P2.54mm_Vertical	HCTL PZ254-2-03-Z-8.5	C2894964	hand/THT
J8	Endstop X	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J9	Endstop Y	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J10	Endstop Z	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
R48	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
C41	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
R49	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
C42	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
R50	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
C43	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
J11	X driver logic	PinSocket_1x08_P2.54mm_Vertical	hanxia HX PM2.54-1x8P	C22438158	hand/THT
J12	X driver power/motor	PinSocket_1x08_P2.54mm_Vertical	hanxia HX PM2.54-1x8P	C22438158	hand/THT
J20	Motor X	JST_PH_B4B-PH-K_1x04_P2.00mm_Vertical	JST B4B-PH-K-S(LF)(SN)	C131334	hand/THT
R51	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
J13	Y driver logic	PinSocket_1x08_P2.54mm_Vertical	hanxia HX PM2.54-1x8P	C22438158	hand/THT
J14	Y driver power/motor	PinSocket_1x08_P2.54mm_Vertical	hanxia HX PM2.54-1x8P	C22438158	hand/THT
J21	Motor Y	JST_PH_B4B-PH-K_1x04_P2.00mm_Vertical	JST B4B-PH-K-S(LF)(SN)	C131334	hand/THT

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
R52	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
J15	Z driver logic	PinSocket_1x08_P2.54mm_Vertical	hanxia HX PM2.54-1x8P	C22438158	hand/THT
J16	Z driver power/motor	PinSocket_1x08_P2.54mm_Vertical	hanxia HX PM2.54-1x8P	C22438158	hand/THT
J22	Motor Z	JST_PH_B4B-PH-K_1x04_P2.00mm_Vertical	JST B4B-PH-K-S(LF)(SN)	C131334	hand/THT
R53	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)

Vibration puck bill of materials

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
U1	ADXL345	LGA-14_3x5mm_P0.8mm_LayoutBorder1x6y	ADXL345BCCZ-RL7	C9667	SMT (JLC)
C1	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
C2	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
C3	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
J3	SPI to controller	PinHeader_2x03_P2.54mm_Vertical	HCTL PZ254-2-03-Z-8.5	C2894964	hand/THT

Bought-in

StepStick driver modules (1-3), NEMA17 motors, 12-28 V supply (24 V recommended) with current limit capability for bring-up, 1x8 female socket strips are on the BOM, endstop switches, 6-wire cable for the puck (under 30 cm), M2.5 screws.

4. Make the boards

1. Upload `utility10/pcb/release/kinetic_ctl/1_fabrication/kinetic_ctl_gerbers.zip` to the PCB fab (2 layers, 1.6 mm, ENIG or lead-free HASL, green mask, white silk; order 2 oz copper if you will drive more than about 1 A per motor phase: the motor traces are 0.4 mm). 2. For assembly upload `2_assembly/kinetic_ctl_JLC_BOM.csv` and `kinetic_ctl_JLC_CPL.csv`; check every polarity in the preview (diodes, electrolytic capacitor, USB-C, ESP32 module pin 1). 3. Through-hole connectors and pin sockets are listed with LCSC numbers but are normally hand-soldered. 4. General rules, tolerances and inspection: `manuals/STANDARD_PROCEDURES.md` sections A and B. Repeat for `kinetic_vib` (1 oz is fine).

5. Program

Build with Arduino-ESP32 core 2.x or PlatformIO (`esp32-s3-devkitc-1`): sources `kinetic/firmware/src/main_kinetic.cpp` plus `kinetic/firmware/lib/*.c`. First flash over USB-C (hold BOOT, tap RESET). **The firmware compiles for the ESP32-S3 (tools/esp32_build.sh, 0 warnings); bring it up on a bench first: it has not run on hardware.**

6. Commissioning (see `kinetic/docs/TEST_PLAN.md` for the full list)

1. Current-limited 24 V, no drivers fitted: TP1 24 V, TP2 5.0 V, TP3 3.3 V. 2. Reverse the supply once with 100 mA limit: nothing conducts. 3. Fit ONE driver with the motor disconnected; set its current limit; power off; connect the motor. 4. `MOVE X=1000` and watch STEP/DIR on a scope. 5. Endstop trip stops motion and needs `CLEAR`. 6. Puck: device ID 0xE5, rest Z = +1 g.

7. Safety and limits

Never plug or unplug a motor with power on (destroys drivers). The TVS and bulk capacitor assume at most 28 V. No ESD protection on USB data. The 3.3 V LDO is limited to 600 mA. The vibration classifier is a demonstration trained on simulated faults: do not use it to decide maintenance without retraining on real recordings. Not safety-rated: it is not an emergency-stop system.